Xiaoyan Wu

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Academic Positions

University of Zurich & ETH Zurich Dec 2024–Present
Psychiatric University Hospital Zurich and Neuroscience Center Zurich Zurich, Switzerland
Postdoctoral Fellow. Supervisor: Philipp Homan

Education Background

Beijing Normal University, School of Psychology Oct 2024
PhD in Cognitive Neuroscience. Supervisor: Chao Liu and Hang Zhang Beijing, China

University of Zurich, Zurich Center for Neuroeconomics Jan 2024
PhD in Neuroeconomics (2 years Joint Training). Supervisor: Christian C. Ruff Zurich, Switzerland

Shenzhen University, Department of Psychology Jun 2020
MSc in Psychology. Supervisor: Yuejia Luo Shenzhen, China

Zhuhai College of Jilin University, Business School Jul 2017
BS in Economics and Business Administration Zhuhai, China

Research Funding

- University of Zurich Postdoc Grant, Zurich, Switzerland. Sep 2026–Aug 2027
  PI, 110,000 CHF
- Anthropic AI for Science Award, California, US. Sep 2026–Mar 2027
  PI, 20,000 USD

Selected Awards

- Best Poster Presentation Award, First Prize (sole awardee) 2026
  20th Alpine Brain Imaging Meeting, Champéry, Switzerland.
- Best PhD Dissertation Award (Top 1%) 2025
  Beijing Normal University, Beijing, China.
- S4SN Travel Award 2024
  Society for Social Neuroscience Annual Meeting, Tsukuba, Japan.
- Outstanding Graduate (Top 1%) 2020
  Shenzhen University, Shenzhen, China.
- National Scholarship (Top 1%) 2018
  Ministry of Education of China.

Selected Publications † Co-first authors. Full list on Google Scholar.

- **Wu, X.**, Rabe, F., … Homan, P. (2026). Disorganization impairs cognitive maps built from visual inputs. ***Schizophrenia Bulletin***, in press. DOI
- **Wu, X.**†, Fu, H.†, Aydogan, G., … & Liu, C. (2026). The self-interest of adolescents overrules cooperation in social dilemmas. ***eLife***, 14, RP106840. DOI (Press release, *eLife*)
- **Wu, X.**†, Ren, X.†, Liu, C., & Zhang, H. (2024). The motive cocktail in altruistic behaviors. ***Nature Computational Science***, 4(9), 659–676. DOI (Cover runner-up of 2024, *Nature Computational Science*)
- Wang, H., **Wu, X.**, … & Qin, S. (2024). Acute stress during witnessing injustice shifts third-party interventions from punishing the perpetrator to helping the victim. ***PLOS Biology***, 22(5), e3002195. DOI (Primer article, *PLOS Biology*)

- **Wu, X.**, Zhu, R., & Liu, C. (2024). A cognitive computational mechanism for mutual cooperation: The roles of positive expectation and social reward. ***Acta Psychologica Sinica***, 56(8), 1001–1015. DOI
- Jin, K., Wu, J., Zhang, R., Zhang, S., **Wu, X.**, Wu, T., Gu, R., & Liu, C. (2024). Observing heroic behavior and its influencing factors in immersive virtual environments. ***Proceedings of the National Academy of Sciences of the United States of America***, 121(17), e2314590121. DOI
- **Wu, X.**†, Feng, C.†, He, Z., Gong, X., Luo, Y. J., & Luo, Y. (2019). Gender-specific effects of vasopressin on human social communication: An ERP study. ***Hormones and Behavior***, 111, 1–9. DOI

Preprints

- **Wu, X.**, & Shi, F. (2026). Large language models underrepresent malice and overreport virtue. ***arXiv***, 2611.01332. DOI
- **Wu, X.**, & Dreher, J. C. (2026). Human-like moral judgments conceal divergent motive attributions in large language models. ***arXiv***, 2609.07353. DOI

Selected Invited Talks

- Neurocomputational components of trust
  Invited by Prof. Jean-Claude Dreher at *Centre national de la recherche scientifique, Lyon* Jul 2026
- From perception to structure: cognitive disorganization is linked to weaker relational knowledge construction
  Invited by Prof. Raymond Dolan at *University College London, London* Jun 2026
- Disorganization impairs cognitive maps built from visual inputs
  Invited by Prof. Seongmin Park at *Virginia Tech, Blacksburg* Feb 2026
- Quantifying language deficits in schizophrenia via neural abstract space mapping
  Invited by Prof. Wolfram Hinzen at *Universitat Pompeu Fabra* Jun 2025
- The "motive cocktail" in altruistic behaviors
  Invited by Prof. Yohsuke Ohtsubo at *University of Tokyo* Mar 2024
- Neurocomputational components of trust
  Invited by Prof. Brian Knutson at *Stanford University* Jan 2024

Selected Conference Presentations

- **Wu, X.**, Rabe, F., . . . , & Homan, P. Disorganization impairs cognitive maps built from visual inputs. *Schizophrenia International Research Society*, Florence, Italy, 2026.
- **Wu, X.**, Rabe, F., . . . , & Homan, P. Disorganization impairs cognitive maps built from visual inputs. *Alpine Brain Imaging Meeting 2026, 20th Anniversary Edition*, Champéry, Switzerland, 2026.
- **Wu, X.**, Rabe, F., . . . , & Homan, P. Cognitive disorganization impairs the ability to construct and use cognitive maps formed from visual perceptual inputs. *8th Burghölzli Psychiatry Meeting*, Zürich, Switzerland, 2025.
- **Wu, X.**, Ren, X., Liu, C., & Zhang, H. The neural signature of the motive cocktail in altruistic behavior. *22nd Annual Meeting of the Society for Neuroeconomics*, Cascais, Portugal, 2024.
- **Wu, X.**, Bürgi, N., & Ruff, C. Cognitive components of trust. *Society for Social Neuroscience Annual Meeting*, Tsukuba, Japan, 2024.

- **Wu, X.**, Ren, X., Liu, C., & Zhang, H. The motive cocktail in third-party punishment and helping. *21st Annual Meeting of the Society for Neuroeconomics*, Vancouver, Canada, 2023.
- **Wu, X.**, Bürgi, N., & Ruff, C. Cognitive components of trust. *21st Annual Meeting of the Society for Neuroeconomics*, Vancouver, Canada, 2023.

Skills

- **AI evaluation and interpretability:** human-LLM behavioral benchmarking; probing and activation steering of open-weight models.
- **Experimental methods:** interactive multi-player, online, and laboratory behavioral paradigms; neuroimaging (fMRI, EEG, MEG) and eye-tracking.
- **Cognitive computational modeling**.
- **Programming:** Python, R, MATLAB, and JavaScript.
- **Languages:** English; Mandarin, Cantonese, and Teochew (native); German (A2).

Supervision and Professional Service

- Thesis supervision: Master's thesis (University of Zurich) and Bachelor's theses (Beijing Normal University, awarded Outstanding Dissertation).
- Member, Education Committee and Global Representation Committee 2026–2029
  Schizophrenia International Research Society
- Reviewer for *eLife*, *Schizophrenia Bulletin*, *Evolutionary Psychological Science*, *Advances in Psychological Science*, *Collabra: Psychology*, *Frontiers in Psychology*, *Frontiers in Behavioral Economics*, and *Frontiers in Genetics*.